

Cosine Similarity Notes

Cosine Similarity

Cosine Similarity is one of the most used methods to find similarity between two vectors.

Formula:

$$\cos(\theta) = (A \cdot B) / (||A|| \times ||B||)$$

Where:

$A \cdot B$ is the dot product of vectors A and B.

$||A||$ is the Euclidean (L2) norm of A.

$||B||$ is the Euclidean (L2) norm of B.

Output Range:

The value lies between -1 and 1.

1 → Exactly similar

0 → No similarity

-1 → Opposite

Using sklearn:

```
from sklearn.metrics.pairwise import cosine_similarity
```

```
A = [[1, 2, 3]]
```

```
B = [[2, 4, 6]]
```

```
print(cosine_similarity(A, B))
```

Row-wise similarity:

```
cosine_similarity(A, B)
```

Column-wise similarity:

```
cosine_similarity(A.T, B.T)
```

(.T is used to transpose the matrix, converting columns into rows)

Manual Calculation:

```
import numpy as np
```

```
A = np.array([
```

```
    [1, 2, 3],
```

```
    [1, 72, 12],
```

```
    [1, 4, 3]
```

```
])
```

```
B = np.array([2, 4, 6])
```

```
cos_sim = np.dot(A, B) / (np.linalg.norm(A, axis=1) * np.linalg.norm(B))
```

```
print(cos_sim)
```

Axis Explanation:

axis=1 → Row-wise operation

axis=0 → Column-wise operation

Key Takeaway:

Cosine similarity measures similarity based on direction (angle), not magnitude.